

Jacob A. Likins

jacobalikins@gmail.com | linkedin.com/in/jacoblikins | jacobal.me | (864) 363-6085

Education

Olin College of Engineering, Needham, MA

Bachelor of Science, Electrical and Computing Engineering, May 2028

- **Relevant Coursework:** Intro Microelectronic Circuits, Signals, Computer Architecture, Software Design, Introduction to Circuits, Linear Algebra, Modeling & Simulation, Engineering Design, Quantitative Engineering Analysis (3D Modeling, Calculus I, II, and III, MATLAB), Probabilistic Design Optimization

South Carolina Governor's School of Mathematics and Science, Hartsville, SC, Governor's High School degree, May 2024

- **Relevant Coursework:** Writing in STEM, Robotics, Engineering Disciplines, AP Chemistry, AP Physics, Electromagnetism, and Mechanics

Software Experience

Olin Electric Motor Sport

Electrical Lead, 7/2025 – *present*, Needham, MA

- Contributes to the design, construction, and testing of an electric vehicle for the Formula SAE Electric competition.
- Leads 24 electrical members in the development of electric vehicle
- Developed a modular firmware architecture using Bazel and custom peripheral libraries for STM32/ARM microcontrollers.

Electrical team, 9/2024 – 7/2025, Needham, MA

- Created PCB boards for battery management, throttle sensing, and pre-charge/discharge
- Integrated regen braking on the car by changing firmware in C to increase endurance on the vehicle
- Programmed the motor controller to work with our CAN protocol

Web Development

Freelancer, 5/2025 – 8/2025, Greenville, SC

- Worked with three separate companies on website development
- Increased the company's marketability

Clemson University International Center for Automotive Research

Intern, 6/2023 – 7/2023, Greenville, SC

- Designed a robot in SolidWorks, then used VEX parts to create a robot
- Programmed the robot's motors and sensors in C++ inside of VS Code
- Tested the robot in Isaac Simulator and compared it to a real-life robot
- Presented at the Governor's Research Colloquium

First Technical Robotics

Team Captain, 9/2022 – 3/2024, Hartsville, SC

- Led team to build a functional robot at the end of the First Tech Challenge season
- Designed a robot and created custom 3-D printed parts with SolidWorks
- Programmed a robot in Java with Android Studios as the IDE
- Designed a workshop to teach members how to use SolidWorks, 3-D printing, and design a robot

Remote Controllable Golf Cart

8/2023 – 4/2024, Hartsville, SC

- Took a Golf Cart and welded an actuator onto the front to control steering
- Used an Arduino Nano to control electronic components and soldered wiring together

Other Experience

Will's Remodeling

General Contracting, 4/2022 – 8/2024, Westminster, SC

- Read schematics to rewire electrical systems in the house

Key Skills

Tools: Bazel (Build Systems), Git, KiCad, SolidWorks, Visual Studio Code (VS code), Oscilloscopes

Software\Firmware Languages: MATLAB, Python, C++, System Verilog, Bash, HTML, CSS, JavaScript, Java

Soft Skills: Teamwork, Leadership, Scrum, Communication, Problem Solving

Other Skills: SMT Soldering, Welding, Wood Cutting, 3-D printing